MILITARY SPECIFICATION

FUZE, ROCKET, M427, MOD 5
LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the quality assurance provisions and special requirements not covered by the drawings, for the fuze designated as Fuze, Rocket, M427, MOD 5.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-46464(MU) - Pellets, Tetryl
MIL-P-60412(MU) - Packaging, Packing and Marking for Shipment of Artillery Type and Rocket Fuzes.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes.

FSC: 1340

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DRAWINGS

ORDNANCE CORPS

9234554 - Fuze, Rocket, M427, MOD 5 (Loading)

PUBLICATIONS

ORDNANCE CORPS

ET 9234554 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Fuze assemblies.-The loaded fuze assembly shall comply with the requirements specified on drawing (dwg.) 9234554 and with all requirements specified in applicable specifications.

3.3 Detail requirements.-The detail requirements of MIL-P-46464 are applicable with the exception of para. 4.2.2.1.

3.4 Booster Functioning.-The fuze assembly (with live booster assembly) when armed and functioned shall detonate high order.

3.5 Ballistic firing.-The fuze shall cause the warhead to detonate high order upon impacting natural ground surface.

3.6 Workmanship.-The loading and assembly operations shall be performed in a thorough, workmanlike manner consistent with the best practices of the industry. All parts and assemblies shall be free of dirt, grease, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Surface coatings shall be continuous except for a few slight scratches not exposing base material. All required markings shall be neat and sharply defined.

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3.7 First Article Inspection.-This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Submission of product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.1.2 Use of simulated defect standards. If automatic radiographic equipment is used, it shall be validated hourly using Design Agency approved standards containing simulated defects and supplied by the contractor. These standards should be representative of defects actually found or anticipated. The contractor shall resupply the item standards when it is determined by the Government that the standards are no longer representative of the product.

4.1.3 Undetected defect standards. Provisions shall be established for the corrective action to be taken where the equipment fails to identify a standard as defective. These provisions shall include a reinspection of all acceptable material subsequent to the last acceptable verification check as a minimum.

4.2 First Article Inspection

4.2.1 Submission.-The contractor shall submit a first article sample consisting of fifty (50) fuze assemblies (with live booster) to be subjected to Booster Functioning. This sample is submitted to a Government approved facility as designated by the contracting officer (see 6.4) for evaluation in accordance with the provisions of 4.2.2. All parts and materials shall be the same as will be used in production and shall be obtained from the same source of supply. Prior to submission, the sample shall be inspected by the contractor to the degree necessary to assure that it conforms to the requirements of the contract.

4.2.2 Inspections to be performed.-Each fuze assembly and component shall be subjected by the Government to any or all of the examinations or tests specified in 4.3.2 and 4.3.3 of this specification and any or all requirements of the applicable drawings.

4.2.3 Rejection of first article sample.-If any fuze, subassembly, or component fails to comply with the applicable requirements the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a fuze, subassembly or component in the sample to comply with any of the specified requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article sample or portion thereof. Until a first article sample is accepted, the contractor is, in no way, authorized by the Government to initiate regular production (including procurement of material by the contractor) unless otherwise directed by the contracting officer.

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4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of loaded fuzes shall contain:

a. One metal parts lot from one manufacturer.

b. Tetryl from not more than one lot or homogeneous blend of lots for each type of pellet, dwg. 8883684 or 8883685.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Explosive, booster (see dwg. 8883685 covering a detail of dwg. 9234554).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Diameter	Gage	01001
102.	Length	Gage	01002
103.	Weight	Balance	01003
Minor:	AQL 1.0 percent		
201.	Cracks or laminations	Visual	01004
202.	Evidence of poor workmanship (see 3.6)	Visual	01005

4.3.2.2 Explosive, lead in (see dwg. 8883684 covering a detail of dwg. 9234554).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Weight	Balance	02001
102.	Diameter	Gage	02002
103.	Length, min	Gage	02003
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.6)	Visual	02004

4.3.2.3 Fuze Assembly, after assembly of lead-in (see dwg. 9234554).

Categories	Defects	Method of Inspection	Code No.
Critical:	None Defined		
Major:	AQL 0.40 percent		
101.	Presence of explosive lead-in	Visual	03001
Minor:	None defined.		

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4.3.2.4 Fuze assembly, prior to crimp (see dwg. 9234554)

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Pellet, pad, or closing disc missing	Visual/Manual	04001
Minor:	Evidence of poor workmanship (see 3.6)	Visual	04002

4.3.2.5 Fuze Assembly (see dwg. 9234554).

Categories	Defects	Method of Inspection	Code No.
Critical: None Defined			
Major:	AQL 0.40 percent		
101.	Length from shoulder to end of booster assembly, max	Gage	05001
102.	Varnish missing from aft end	Visual	05002
103.	Varnish where not permitted	Visual	05003
Minor:	AQL 1.50 percent		
201.	Marking missing, misleading or unidentifiable	Visual	05004
202.	Protective finish inadequate	Visual	05005
	(see 3.6)	Visual	05006

4.3.3 Testing

4.3.3.1 Composition of Tetryl (see dwgs. 8883684 and 8883685) Major defect - Code No. 01006 or 02005.-The Tetryl composition of each batch or lot shall be determined in accordance with 4.4.1.

4.3.3.2 Radiographic inspection for armed fuzes - Critical defect - Code No. 06001.-Each fuze shall be radiographically inspected for an armed or partially armed condition. Fuzes found to have either of these defects shall be removed from the lot, held for engineering evaluation and the lot shall be rejected. This inspection shall be performed as specified in 4.4.2. Defect standards for automatic equipment verification shall be in accordance with 4.1.2. Failure to detect defect standards shall put in effect procedures in accordance with 4.1.3.

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4.3.3.3 Radiographic inspection for missing or broken components - Major Defect - Code No. 06002.-Each fuze shall be radiographically inspected for broken or missing internal components. Fuzes found to have this type of defect shall be removed from the lot. This inspection shall be performed as specified in 4.4.2. Defect standards for automatic equipment verification shall be in accordance with 4.1.2. Failure to detect defect standards shall put in effect procedures in accordance with 4.1.3.

4.3.3.4 Booster Functioning (see 3.4) - Major defect - Code No. 07001.-Beginning with the first production lot, fifty (50) complete fuzes with live boosters shall be selected and tested from each lot. If two or more fuzes fail to detonate high order the lot shall be rejected. The fifty (50) samples shall all be fired regardless of the failure rate. This test shall be performed as specified in 4.4.3.

4.3.3.5 Ballistic Test (see 3.5) - Major defect - Code No. 08001.-This test shall be performed on warhead-fuze combination lots. All sample assemblies shall be numbered and x-rayed for information only. The assemblies shall be shipped to the proving grounds, retaining the identification numbers and attached to motors for ballistic test. A random sample of twenty-four (24) loaded fuzed warheads shall be selected from each lot. Should two or more rounds fail to detonate high order, the lot shall be rejected. Should one round fail to detonate high order, an additional twenty-four (24) samples shall be tested. Should one or more of these additional twenty-four (24) samples fail to detonate high order, the lot shall be rejected.

This test shall be performed as specified in paragraph 4.4.4.

4.3.4 Inspection Equipment.-Equipment Tabulation Number ET-9234554 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

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4.4 Test methods and procedures.

4.4.1 Composition of tetryl.-Tetryl composition shall be determined in accordance with MIL-P-46464.

4.4.2 Radiographic Inspection.-Fuzes shall be radiographed perpendicular to the longitudinal axis. Inspection for armed or partially armed fuzes and inspection for broken or missing components may be made simultaneously.

4.4.3 Booster functioning.-The assembled fuze shall be armed on a centrifuge by application of 40.0 gravities "g's" maximum for a minimum of three (3) seconds which simulates a rocket launching. Radiographic inspection shall be utilized to verify that each unit is armed and has no defective or missing internal parts. For functioning, each fuze shall be positioned vertically on top of an aluminum witness block. Each fuze shall be functioned by dropping a fifty (50) to fifty-one (51) pound weight from a height of 5.0 feet to 5.5 feet onto the nose of the fuze. The weight shall be dropped only once. High order detonation shall be determined by a 0.25 inch minimum dent in the cylindrical witness block (1.5 inch min. height by 2.0 inch min. diameter of 2011 - T3 aluminum). WARNING: This test is dangerous and must be performed behind a suitable barrier.

4.4.4 Ballistic test.-The complete fuze with warhead and rocket motor shall be fired from a launcher at a minimum height of 35 feet to impact the ground at a minimum range of 500 yards. The launch tower shall use a Standard 2.75 Inch Rocket Launcher (or equivalent) which has an inside diameter of 2.822 $\pm$ .010 inches and a detent to restrain the rocket until a thrust of 200 $\pm$ 10 pounds is developed. Each launch fixture shall be inspected for compliance with the required diameter and release force a minimum of once per month. Defective launcher components shall be replaced as needed. A high order warhead function which occurs on second impact shall be considered a valid test. Acceptable high order warhead functions may occur short of the intended 500 yard range since M427 fuzes can be armed at a minimum range of 150 yards. Duds at ranges below 500 yards shall be discounted from the test results. Unobserved impacts shall be indicated as such and may be discounted from the test results if sufficient information is not available to make a function determination. Ballistic reports of samples which contain more than one fuze lot shall include round by round data of which fuze lot each round represents.

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5. PREPARATION FOR DELIVERY

5.1 Packaging and marking.-The fuze assemblies shall be packaged and marked in accordance with MIL-P-60412.

5.2 Packing and marking.-The packaged fuzes shall be packed and marked in accordance with MIL-P-60412.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.

b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. They should also be cited as references on drawings of equipment designs submitted by the contractor (see 6.3.5).

6.3 Inspection equipment. - The contractor shall design inspection equipment as required by the referenced inspection Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5. Equipment Lists for the M423/M427 fuze, which are applicable to the M427 Mod 5 fuze, may be employed for the inspection of similar defects.

6.3.1 Inspection equipment lists. - Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.2 Inspection equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs. - Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs. - Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.5 Submission of contractor designs. - All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND9. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

6.4 Submission of First Article Samples.-Instructions, as to the location for the evaluation of the first article sample shall be obtained from the contracting officer. Upon receipt of such request, the contracting officer shall advise Picatinny Arsenal and instructions will be issued accordingly. All inquiries should be forwarded to: Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND9.

6.5 Government furnished parts

6.5.1 Ammunition boxes.-The procuring agency shall furnish the contractor sufficient ammunition Boxes, M2A1 to satisfy the shipping requirements for the contract. The top and bottom supports of dwg. 8882443 will not be furnished.

6.6 Submission of test data. In addition to normal distribution of test data, two copies will be forwarded to Picatinny Arsenal, Dover, New Jersey, ATTN: SMUPA-ND9 and SMUPA-DJ3.

6.7 Intended use.-It is intended that the item of this specification will be used in the 2.75 Inch Rocket.

Custodian:
Army-MU

Preparing activity:
Army-MU

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